

STROBE Statement—Checklist for cohort studies

Completed for: PhysioRisk: an interpretable survival-modeling framework for separating demographic and physiological mortality risk

This checklist maps the official STROBE cohort-study items to the submitted manuscript. It does not add manuscript content. PARTIAL means the item is not fully or explicitly reported.

Section	Item	Recommendation	Where reported
Title and abstract	1(a)	Indicate the study's design with a commonly used term in the title or the abstract	Abstract and Methods (Study population, mortality linkage, and temporal split): mortality-linked NHANES 1999–2016 data with development and internal temporal-validation cohorts identify the observational temporal-validation design.
	1(b)	Provide in the abstract an informative and balanced summary of what was done and what was found	Abstract (structured Background, Methods, Results, Conclusions).
Introduction	2	Explain the scientific background and rationale for the investigation being reported	Introduction, paragraphs 1–7.
	3	State specific objectives, including any prespecified hypotheses	Introduction, final paragraph (four stated questions and study objective).
Methods	4	Present key elements of study design early in the paper	Methods: Study population, mortality linkage, and temporal split, paragraphs 1–3.
	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	Methods: Study population, mortality linkage, and temporal split (NHANES cycles 1999–2016; 2019 linked mortality files; follow-up origin and censoring).
	6(a)	Give eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up	Methods: Study population, mortality linkage, and temporal split, paragraphs 1–2; Missing data and cohort selection, paragraphs 1–2; Figure 1.
	6(b)	For matched studies, give matching criteria and number of exposed and unexposed	NOT APPLICABLE — no matched cohort design.
	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	Methods: Study population (all-cause mortality); DemoRisk; PhysioRisk; Calibrated total risk; Comparator; Validation and statistical analysis. No diagnostic criteria were applicable.
	8	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability if more than one group	PARTIAL — Methods: Study population and mortality linkage; PhysioRisk; Comparator. NHANES and mortality-file sources are identified, but assay-level measurement detail is not explicitly reproduced in the manuscript.
	9	Describe any efforts to address potential sources of bias	Methods: temporal split, development-only fitting, unchanged validation application, Missing data and cohort selection; Limitations, paragraphs 1–5.
	10	Explain how the study size was arrived at	No formal sample-size calculation was performed. The analytic sample was determined by mortality eligibility and complete availability of the required biomarkers (Methods: Study population, mortality linkage, and temporal split; Missing data and cohort selection; Figure 1).
	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	Methods: DemoRisk; PhysioRisk; Calibrated total risk; Validation and statistical analysis (splines, residualization, standardization, score scaling, calibration deciles).
	12(a)	Describe all statistical methods, including those used to control for confounding	Methods: DemoRisk through Validation and statistical analysis. Age and sex handling is specified; the work is predictive rather than a causal confounding analysis.
	12(b)	Describe any methods used to examine subgroups and interactions	NOT APPLICABLE — no subgroup analyses were performed. Time-interaction diagnostics, which are not subgroup analyses, are reported in Methods: Validation and statistical analysis.

	12(c)	Explain how missing data were addressed	Methods: Missing data and cohort selection, paragraphs 1–2; Supplementary Tables S6–S8.
	12(d)	If applicable, explain how loss to follow-up was addressed	Methods: Study population and mortality linkage (administrative censoring); Validation and statistical analysis (IPCW and censoring distributions).
	12(e)	Describe any sensitivity analyses	Methods: Missing data and cohort selection; Validation and statistical analysis; Supplementary Tables S5 and S8.
Results	13(a)	Report numbers of individuals at each stage of study	Results: Cohort construction and missingness; Figure 1; Table 1.
	13(b)	Give reasons for non-participation at each stage	Methods and Results (Cohort construction and selection) report exclusion through mortality ineligibility and incomplete required biomarker data; Figure 1 shows cohort flow; Supplementary Tables S6 and S7 report missingness and included/excluded participants.
	13(c)	Consider use of a flow diagram	Figure 1 (cohort flow).
	14(a)	Give characteristics of study participants and information on exposures and potential confounders; indicate missing data for each variable	Table 1; Results: Cohort construction and missingness; Supplementary Tables S6 and S7. Detailed missingness is supplementary.
	14(b)	Summarise follow-up time	Table 1 (follow-up in months); Methods: Validation and statistical analysis (reverse Kaplan–Meier support); Supplementary Table S7.
	15	Report numbers of outcome events or summary measures over time	Methods and Results: cohort deaths; Table 1; fixed-horizon mortality in Results and Supplementary Table S4.
	16(a)	Give unadjusted estimates and, if applicable, confounder-adjusted estimates and precision. Make clear which confounders were adjusted for and why	NOT APPLICABLE to separate unadjusted causal-effect estimates because the study evaluates prediction and temporal validation, not a causal effect. Results and Table 2 report predictive hazard ratios, discrimination estimates, and 95% confidence intervals; Methods specifies age/sex handling and model construction.
	16(b)	Report category boundaries when continuous variables were categorized	NOT APPLICABLE to the primary continuous scores; calibration used risk deciles as stated in Methods.
	16(c)	If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	Results: Calibration and follow-up support; 5- and 8-year predicted and observed mortality; Supplementary Table S4.
	17	Report other analyses done—e.g., subgroup and sensitivity analyses	Results: Proportional hazards; Calibration and follow-up support; robustness analyses; Supplementary Tables S2–S5 and S8. No subgroup analyses were performed.
Discussion	18	Summarise key results with reference to study objectives	Discussion, opening paragraph and Interpretation of the decomposition.
	19	Discuss limitations, taking into account sources of potential bias or imprecision	Limitations, all five paragraphs.
	20	Give a cautious overall interpretation considering objectives, limitations, multiplicity, similar studies, and other evidence	Discussion, especially Relationship to biological-age and acceleration approaches; Limitations; Conclusions.
	21	Discuss the generalisability (external validity) of the study results	Limitations, paragraphs 1–2; Conclusions (external validation required).
Other information	22	Give the source of funding and the role of the funders	Statements and Declarations — Funding: no external funding; therefore no funder role.

Note: An item being mapped does not imply that every requested detail is fully reported. Gaps are explicitly labelled PARTIAL, NOT APPLICABLE, or NOT REPORTED.